

Financial Time Series

Monday May 3, 2021

Monday TA Session #4

X_t = mean model + noise model
(ARMA)
log returns
of MRP
 $\{a_t\}$
 $WN(0, \sigma^2)$

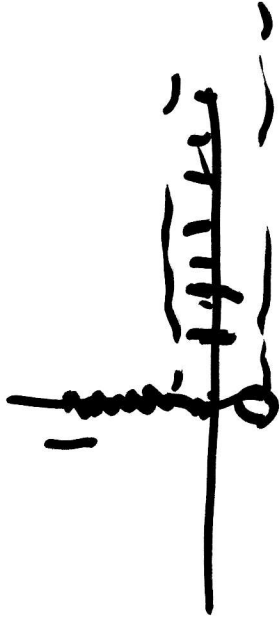
~~ARMA~~ AIRMA-ORDER-SELECT-IC ()
AIC, BIC

[4, 27] for both AIC, BIC

Model = ARMA.fit(logret, [4, 27], ...)

Summarize Model - check residuals
Are there ARCH effects?

Model-resid, Look at ACF, PACF plots
to see if there (is/are) any
unmodeled temporal behavior
(i.e. are the residuals reasonably
like white noise??)



Then check for ARCH effects.

Engle test on squared residuals
Ljung-Box test on squared residuals

If there are ARCH effects, then fit
ARCH/GARCH model

Look at ~~test~~ standardized residuals
from the ARCH/GARCH fit

Run Engle test on squared residuals
to check if there are any unmodeled
ARCH effects (p value = .99.)

Cointegration (Lecture 12 slides 22-23
why not apply to diff(data)
?)

Cointegration - $\{X_1(t), X_2(t)\}$

Not stationary

But X_1, X_2 are said to be cointegrated if $\exists (a, b)$
for which $aX_1(t) + bX_2(t) = Y(t)$

$\{Y(t)\}$ is ~~not~~ stationary and (a, b) is called
a cointegration vector

③ Now - we first reduce both to $I(1)$ by
appropriate differencing.

Trace statistic for $H_0: R=0$

131.2758... $H_1: R>0$

> 35.4128

$\Rightarrow$ Reject H_0 at $\alpha=0.01$ level

$H_0: R=1$

test stat (Trace) = 10.2835...

$H_1: R>1$

Accept at $\alpha=0.10$ level

(and at 0.05, 0.01)

Conclusion is $R=1$

There is one cointegrating vector.